Robert Laughlin

DATA ANALYST

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Open to remote work and relocation to the Des Moines area

PROFILE

Entry-level data analyst with strong SQL, Harvard's CS50 SQL certificate, and verified statistics and calculus certificates (Stanford, MITx, DeepLearning.AI). Built a public BigQuery pipeline and dbt star schema for 403,264 Polk County, Iowa residential sales (1990-2026). Writes analytical SQL with window functions, conditional aggregation, and as-of joins, and checks row grain and join fan-out.

CORE SKILLS

SQL: window functions, CTEs, conditional aggregation, as-of joins, indexes, star schemas, join-cardinality and fan-out checks; BigQuery, SQLite, MySQL

Python: pandas, NumPy, Matplotlib, Seaborn, Jupyter

Statistics: descriptive statistics, distributions, missingness and outlier audits, probability (Bayes, expectation and variance, LLN/CLT)

Tools: Excel, Git and GitHub, dbt, Docker, Linux

SELECTED PROJECTS

Polk County Residential Sales Pipeline | **2026**

Python, SQL, BigQuery, dbt, Terraform, Docker | GitHub

- Loaded 403,264 Polk County Assessor residential sale records (1990-2026) into BigQuery with a Dockerized Python job that retries failed downloads and uploads only new or changed yearly files (MD5 comparison); Terraform provisions the Cloud Storage bucket and datasets.
- Modeled the sales in dbt as a star schema for sale-price analysis: sale-level fact tables, parcel, building, date, and code dimensions, and bridge tables for sales that cover several parcels. Building records are versioned, so each sale sees the house as it was when it sold.
- Wrote 205 dbt tests on keys, relationships, and value ranges. Warning-level tests flag source-data problems, such as 249 sales whose total assessment isn't land plus building, and keep the failing rows in audit tables.

Analytical SQL | **2026**

SQL, SQLite, MySQL, Harvard CS50 SQL

- Completed Harvard's CS50 Introduction to Databases with SQL (certificate): every problem set, and a final project that loads a 2,000-sale sample of the Polk County star schema into SQLite with foreign keys, views, and indexes.
- Fixed a fan-out bug by aggregating costs to one row per sale before joining, and wrote as-of joins that match each sale to the assessment published on or before its sale date (ROW_NUMBER over partitions).

Exploratory Data Analysis | **2026**

Python, pandas, Seaborn, Matplotlib

- Profiled the Palmer Penguins dataset (344 rows): group summaries (mean, median, IQR), missingness by group (9.6% vs. 0.8% across islands), and an outlier and record-error audit comparing three correction scenarios.

Dispatch Assignment Tool & KNN Demo | **2021**

Python, NumPy, nearest-neighbor methods | GitHub Demo

- Built a nearest-neighbor tool that a California state field-assessment team used twice a year to group house locations into assignments for about 10-20 field workers; published a separate vectorized KNN demo.

EDUCATION & CERTIFICATIONS

City University of Seattle | **Coursework toward B.S. in Data Science | 2019-2021**

100 quarter credits. Coursework: foundational statistics, statistical computing, calculus, discrete mathematics, database technologies, and computer science.

Certifications:

- CS50's Introduction to Databases with SQL, Harvard University (2026)
- Introduction to Statistics, Stanford University via Coursera (2025)
- Probability & Statistics for Machine Learning & Data Science, DeepLearning.AI via Coursera (2025)
- Google Project Management Professional Certificate, 6 courses, Google via Coursera (2025)
- Calculus: MITx 18.01.1x and 18.02.1x, verified, edX (2020-2021); Mathematics for Machine Learning: Multivariate Calculus, Imperial College London (2021)
- Deep Learning Specialization, 5 courses, DeepLearning.AI via Coursera (2024)

Self-study: MIT OCW 6.041 Probabilistic Systems Analysis (2026), with worked problem sets.

Current focus: Polk County sale-price analysis in SQL, a BI dashboard, and statistical inference.